

Elaine Yang

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SUMMARY

Analytical professional with experience in data science, investment analytics, and market analytics, skilled in **Python, SQL, Tableau, and ML**. Proven track record building scalable **ETL pipelines**, developing segmentation and recommendation systems, and optimizing engagement through **A/B testing**. Adept at delivering actionable insights in technology and financial sectors to drive business impact.

EDUCATION

Columbia University

M.S., Business Analytics

Aug 2023 - Feb 2025

Relevant Courses: Asset Allocation, Risk Management, Data Analytics, Marketing Analytics, Optimization Models and Methods

University of Illinois at Urbana-Champaign

B.S., Accounting, Information Systems (GPA: 3.9/4.0); James Scholar Highest Honor (Top 5% of college); Accountancy Scholarship

Aug 2019 - May 2023

Relevant Courses: Corporate Finance, Accounting Control System, Info Sys Analysis & Design, Advanced Financial Reporting

WORK EXPERIENCES

OAG Technology | NY, NY

Data Scientist Intern

May 2025 - Present

- Built and maintained a modular **ETL pipeline** (Airflow, BigQuery, Python, SQL) to process ~**80k** monthly creator events; engineered behavioral features for segmentation, churn modeling, and personalization.
- Applied **k-means clustering** to identify creator personas across onboarding/publishing stages, enabling targeted lifecycle interventions, experimentation, and re-engagement strategies.
- Conducted **churn analysis** on new creators using **logistic regression** and survival modeling; identified early engagement thresholds triggering automated in-product nudges for high-risk users.
- Built and deployed a personalized workflow recommendation pipeline using **collaborative filtering** and **LightGBM** re-ranking.
- Designed and ran **A/B and multivariate experiments** on onboarding flows, content templates, and CTAs, and partnered with product/design teams to translate findings into segmentation-driven strategies, resulting in an **18% increase in publish completion** and a **35% lift in re-engagement** from targeted notifications/prompts.

Data Analyst Intern

Feb 2025 – May 2025

- Collaborated with Marketing to enhance campaign targeting by applying **uplift modeling**, overcoming inefficiencies of traditional response models. Distinguished persuadable users, improving resource allocation and acquisition efficiency across email, push, and digital advertising channels.
- Designed and implemented an uplift modeling framework integrating behavioral, demographic, and campaign-interaction data. Produced data-driven targeting strategies that maximized incremental acquisitions while supporting decision-making across multi-channel marketing initiatives.
- Engineered scalable pipelines leveraging **T-Learner, S-Learner, and Uplift Random Forest algorithms**. Enriched models with channel-specific and temporal engagement features; developed an interactive Tableau simulator for **budget optimization** and **ROI** visualization.
- Delivered measurable impact: targeting reduced cost-per-incremental-acquisition by **22%** and improved conversion uplift by **18%** in simulations. Simulator adoption by three sub-teams enhanced budget allocation accuracy and elevated overall marketing efficiency in Q2.

China International Capital Corporation | Data Analyst, Investment | Beijing, CN

Jun 2024 - Aug 2024

- Collaborated with Smart Advisory to build a **portfolio recommendation** engine for high-net-worth clients, part of CICC's core wealth management digital transformation.
- Built an **ETL pipeline (Python, SQL)** to process **200K+** client records from trading, risk profiling, and holdings systems; engineered features like liquidity preference, diversification index, and max drawdown.
- Conducted client segmentation via **k-means clustering** and **PCA** on liquidity, risk tolerance, and portfolio structure; informed tailored portfolio templates and hedging strategies for client tiers.
- Proposed tier-based strategies using Mean-Variance Optimization and Risk Parity; backtested on five-year rolling data and designed a Mass Affluent portfolio with hybrid MVO + Risk Parity, achieving **10.4% annualized return** and **Sharpe ratio >1.2** in simulation.
- Built **Tableau dashboards** to show performance and volatility by client segment and strategy, aiding product team in model evaluation.

TZP Group | Capstone Data Analyst, Private Equity Underwriting | NY, NY

Jan 2024 - May 2024

- Aggregated and standardized financial data for **50+ portfolio** and related companies in Python; normalized revenue, EBITDA, margins, Opex, and free cash flow for cross-company benchmarking and YoY growth reporting.
- Performed correlation studies and **PCA** on 14-dimension executive talent assessments; identified 3 factors most predictive of portfolio company outperformance for integration into investment memos and management updates.
- Developed and validated **5 Random Forest** models to classify out-/under-performers versus base-case projections; achieved **83%** test accuracy and highlighted drivers such as sector exposure and deal duration.
- Built **10+ dashboards** and heatmaps visualizing sector performance, macro trends, and deal source effectiveness; delivered findings in quarterly portfolio reviews and strategic planning sessions for senior leadership.

- Conducted bottom-up financial analysis on leading food and beverage companies (*Yili*), including full income statement and cash flow forecasting. Built three-statement models in Excel to project revenue growth, margin trends, and capital efficiency under multiple macro and pricing scenarios.
- Performed valuation using DCF and relative methods (P/E, EV/EBITDA), adjusting for sector-specific risks and market dynamics.
- Participated in the development of **factor models** for the food and beverage industry. Built a cross-sectional factor database using WIND data; extracted, cleaned, and standardized data via **SQL**. Optimized query logic (JOIN sequence, window functions, CTEs), reducing daily data refresh time by **70%**.
- Applied **NLP** techniques to analyze brokerage reports, annual filings, and financial news. Constructed textual sentiment factors using **TF-IDF** and **FinBERT** embeddings to convert unstructured information into structured predictors, improving the model's forward-looking return predictability and robustness.
- Conducted univariate portfolio sorts and calculated **Information Coefficient (IC)** and **Information Ratio (IR)** to evaluate the predictive power and stability of individual factors.

SKILLS

- **Marketing Related:** A/B Test, Google Analytics, CRM, Conversion Lift Testing, Marketing Mix Modeling
- **Analysis & Visualization:** Tableau, Power BI, Looker Studio, VBA, Python, R, Excel, APIs, Alteryx
- **Database:** Snowflake, BigQuery, SQL Server, Oracle SQL, AWS RDS, NoSQL, MongoDB, Apache Airflow
- **Statistics & Machine Learning:** Causal Inference, Sentiment Analysis, Time Series Forecasting, Boosting Algorithms, Advanced Tree Models, Natural Language Processing